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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 08

Title: Marketing Campaign Analytics Dashboard

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1. Problem Statement

You are a Marketing Analyst in a digital marketing agency. Multiple marketing campaigns have been conducted through different channels, campaign types, companies, and locations. Management wants to evaluate campaign performance based on impressions, clicks, conversion rate, engagement, and ROI. The objective is to develop an interactive Marketing Campaign Analytics dashboard in Power BI that helps management understand campaign performance, compare different marketing channels, identify trends, and make better marketing decisions.

2. Methodology

Step 1: Data Collection

- Collected marketing campaign data containing campaign details and performance metrics.
- The dataset includes fields such as Date, Campaign Type, Channel Used, Company, Location, Impressions, Clicks, Conversion Rate, ROI, and Engagement.
- The data was selected to analyze campaign reach, customer interaction, conversions, and marketing returns.

Step 2: Data Import

- Opened **Power BI Desktop**.
- Imported the marketing campaign dataset.
- Loaded the data into Power BI.
- Verified that all required columns were available.

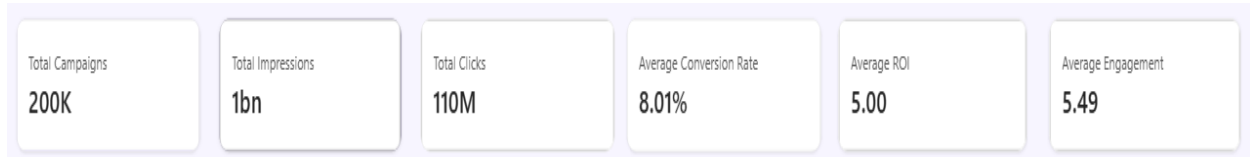
Step 3: Data Cleaning and Transformation

- Checked for missing and null values.
- Checked for duplicate records.
- Verified the data types of all columns.
- Ensured numerical fields such as impressions, clicks, conversion rate, ROI, and engagement were correctly formatted.
- Ensured categorical fields such as campaign type, channel, company, and location were correctly formatted.

Step 4: Dashboard Creation

- Created KPI cards to show important campaign metrics.
- Created charts to analyze ROI, conversion rate, impressions, clicks, and marketing channels.
- Added interactive filters for Date, Campaign Type, Channel Used, Company, and Location.

3. KPI Cards Analysis



1. Total Campaigns

- The dashboard shows approximately 200K campaigns.
- It represents the overall number of campaigns included in the analysis.
- This helps management understand the scale of marketing activities.

2. Total Impressions

- The dashboard shows approximately 1 Billion impressions.
- Impressions represent the total number of times the campaigns were displayed to audiences.
- It helps measure the overall reach of the campaigns.

3. Total Clicks

- The dashboard shows approximately 110 Million clicks.
- Clicks indicate audience interaction with the campaigns.
- A higher number of clicks indicates stronger audience response.

4. Average Conversion Rate

- The average conversion rate is 8.01%.
- It shows the average percentage of users who completed the desired action after interacting with campaigns.
- This KPI helps evaluate campaign effectiveness.

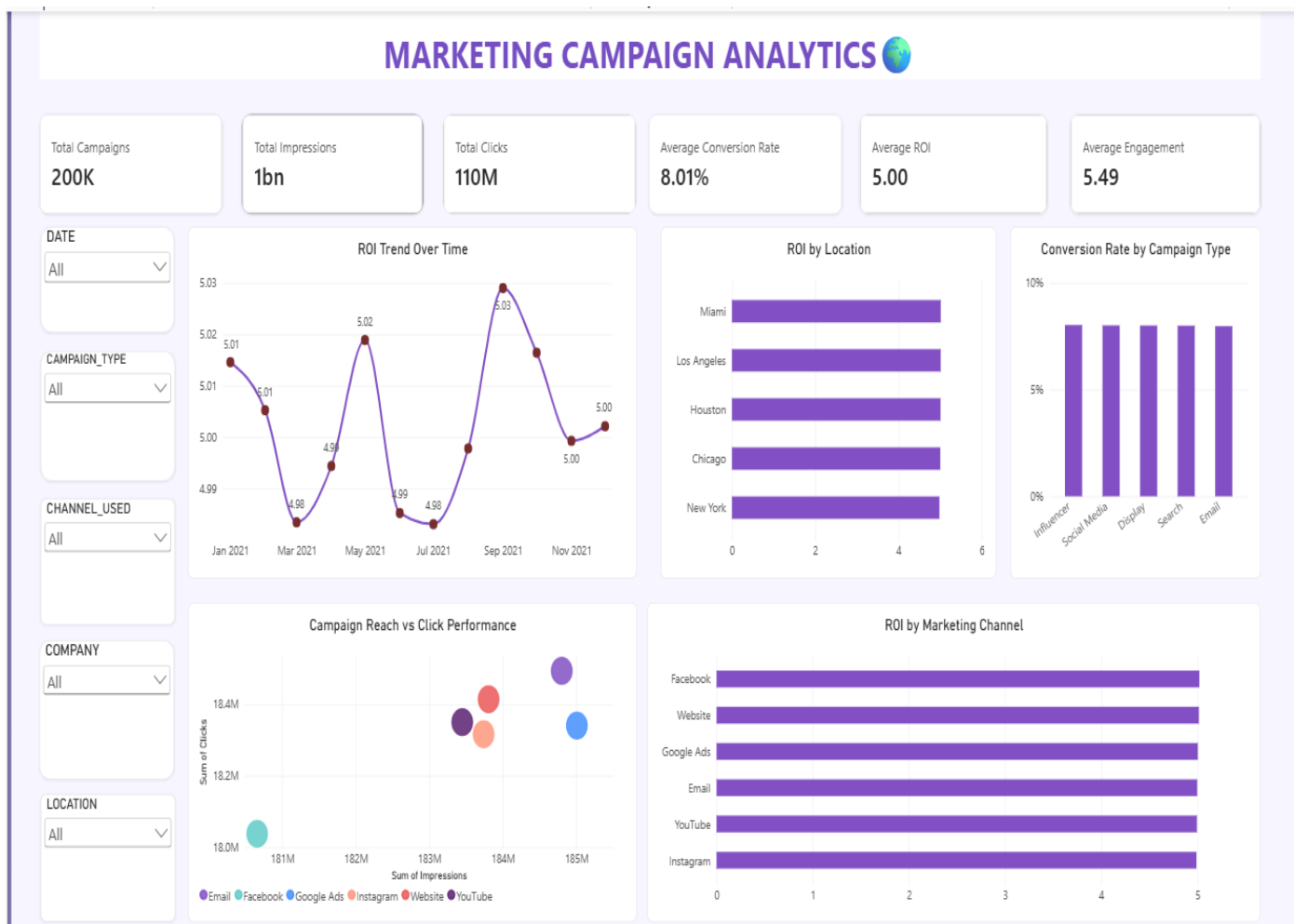
5. Average ROI

- The average ROI is 5.00.
- It indicates the overall return generated from the marketing campaigns.
- Management can use this KPI to evaluate marketing efficiency.

6. Average Engagement

- The average engagement is 5.49.
- It represents the overall level of audience interaction with the campaigns.
- It helps understand how effectively campaigns engage users.

4. Visualization Analysis



1. ROI Trend Over Time

Purpose: Shows how ROI changes over time.

Observation:

- ROI remains close to **5.00** throughout the displayed period.
- The highest visible ROI is approximately **5.03 in September 2021**.
- Lower values of around **4.98** can be seen in some months.

Business Insight:

- Overall campaign ROI is relatively stable.
- Management can investigate months with lower ROI to understand changes in campaign performance.

2. ROI by Location

Purpose: Compares ROI across different locations.

Observation:

- Miami, Los Angeles, Houston, Chicago, and New York have very similar ROI values.
- The ROI values are close to **5.00**.

Business Insight:

- There is no major ROI difference between the displayed locations.
- Location should be analyzed together with clicks, conversions, and engagement before making marketing decisions.

3. Conversion Rate by Campaign Type

Purpose: Compares conversion rates across campaign types.

Observation:

- Influencer, Social Media, Display, Search, and Email campaigns have conversion rates close to **8%**.
- The difference between the campaign types is small.

Business Insight:

- No campaign type has a significantly higher conversion rate in the current dashboard.
- Management should consider ROI, clicks, and engagement along with conversion rate.

4. Campaign Reach vs Click Performance

Purpose: Compares campaign impressions with clicks.

Observation:

- The displayed campaigns have approximately 180M–185M impressions.
- Click performance is around 18M–18.5M clicks.
- Different channels show slightly different click performance even with similar impressions.

Business Insight:

- Higher impressions do not always result in proportionally higher clicks.
- Management should identify channels that convert high reach into better user interaction.

5. ROI by Marketing Channel

Purpose: Compares ROI across different marketing channels.

Observation:

- Facebook, Website, Google Ads, Email, YouTube, and Instagram show ROI values close to **5.00**.
- The difference between channels is very small.

Business Insight:

- No major ROI difference is visible between the displayed channels.
- Other metrics such as clicks, conversions, and engagement should be considered when allocating the marketing budget.

6. Interactive Filters

The dashboard contains filters for:

- **Date**
- **Campaign Type**
- **Channel Used**
- **Company**
- **Location**

These filters allow management to analyze specific campaign segments and compare performance based on different conditions.

5. Overall Insights

- The dashboard contains approximately 200K campaigns.
- The campaigns generated approximately 1 Billion impressions and 110 Million clicks.

- The overall average conversion rate is 8.01%.
- The average ROI is 5.00, showing relatively stable ROI performance.
- The average engagement score is 5.49.
- ROI across the displayed locations is very similar.
- Conversion rates across different campaign types are also close to each other.
- Marketing channels show similar ROI values in the current dashboard.
- Campaigns with similar impressions can still have different click performance.
- Interactive filters make it easier to analyze campaign performance by date, campaign type, channel, company, and location.
- Combining impressions, clicks, conversions, ROI, and engagement provides a better understanding of overall campaign performance.

6. Conclusion

The Marketing Campaign Analytics Dashboard provides an interactive and simple way to evaluate marketing campaign performance. It summarizes important metrics such as campaigns, impressions, clicks, conversion rate, ROI, and engagement through KPI cards and visualizations. The dashboard also helps compare campaign performance across different locations, campaign types, and marketing channels. Interactive filters allow management to perform detailed analysis based on specific requirements. Overall, the dashboard can help the marketing team identify campaign trends, understand audience response, evaluate marketing effectiveness, and make better decisions about future campaigns and budget allocation.

GITHUB LINK: https://github.com/lohithslohiths110405-ui/Data-Analytics-for-Business/tree/main/Marketing_Campaign_Analytics_dashboard